

Project Vows

REST API Documentation

Wedding invitation & WhatsApp RSVP backend

Prepared for the Front-End Team

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Overview

Project Vows is the backend for a WhatsApp-based wedding invitation system. Couples (events) import a guest list via CSV; guests RSVP over WhatsApp and confirmed guests receive a QR code used for check-in at the Holy Matrimony and/or Reception. This document describes every HTTP endpoint the front end will call.

Base URL

```
http://localhost:8080      # local development
https://vows.id           # production (example)
```

All endpoints below are prefixed with **/api** unless noted (the health check lives at the root).

Response envelope

Every JSON response uses the same envelope. **status** is always present; the other fields appear depending on the outcome.

```
{
  "status": "success" | "error",
  "message": "human readable message", // optional
  "data":   { ... } | [ ... ],         // present on success
  "error":  "machine_readable_code"   // present on error
}
```

Front-end tip: branch on the top-level status field, then read data on success or error on failure. The error code is stable and safe to switch on; message is for display/debugging and may change.

HTTP status codes

Status	Meaning
200 OK	Successful read or processed batch operation
201 Created	Resource created (POST /api/events)
400 Bad Request	Malformed body / failed validation / bad CSV
403 Forbidden	Webhook verification token mismatch
404 Not Found	Event / invitation / QR token not found
409 Conflict	Duplicate (e.g. already checked in, duplicate event tag)
422 Unprocessable	Check-in business-rule failure (not attending, etc.)
500 Server Error	Unexpected server error

Field reference (enums)

These string values appear on the invitation object and in filters. Treat them as fixed string constants.

invitation_status

Value	Meaning
imported	Default. Guest imported from CSV, invitation not yet sent
invitation_sent	Initial invitation message sent successfully
invitation_failed	Send attempt failed

rsvp_status

Value	Meaning
not_answered	Default. Guest has not responded yet
attending	Guest confirmed attendance
not_attending	Guest declined

event_choice (nullable)

Value	Meaning
null	Not chosen yet
holy_matrimony	Will attend Holy Matrimony only
reception	Will attend Reception only
both	Will attend both events

qr_status

Value	Meaning
not_generated	Default. No QR generated yet
generated	QR generated & uploaded to WhatsApp media
sent	QR delivered to the guest
send_failed	QR send attempt failed

gift_interest

Value	Meaning
not_asked	Default
yes	Guest wants gift / bank details
no	Guest not interested

event_type (check-in)

Value	Meaning
holy_matrimony	Check-in at the Holy Matrimony
reception	Check-in at the Reception

Events

1. Create event

POST /api/events

Create a wedding/couple record. The **tag** must be unique and is later used as the key for CSV import and all batch operations.

REQUEST BODY

Field	Type	Required	Description
tag	string	yes	Unique slug, e.g. "stanley-arum"
couple_name	string	yes	Display name, e.g. "Stanley & Arum"
holy_matrimony_date	datetime	no	RFC3339, e.g. 2026-06-20T10:00:00+07:00
holy_matrimony_location	string	no	Free text
reception_date	datetime	no	RFC3339
reception_location	string	no	Free text
gift_address	string	no	Free text
bank_account	string	no	Free text

REQUEST EXAMPLE

```
{
  "tag": "stanley-arum",
  "couple_name": "Stanley & Arum",
  "holy_matrimony_date": "2026-06-20T10:00:00+07:00",
  "holy_matrimony_location": "Gereja ABC",
  "reception_date": "2026-06-20T18:00:00+07:00",
  "reception_location": "Ballroom XYZ",
  "gift_address": "Alamat hadiah",
  "bank_account": "BCA 123456789 a/n Stanley"
}
```

RESPONSE EXAMPLE

```
// 201 Created
{
  "status": "success",
  "message": "Event created",
  "data": {
    "id": 1,
    "tag": "stanley-arum",
    "couple_name": "Stanley & Arum",
    "holy_matrimony_date": "2026-06-20T10:00:00+07:00",
    "reception_date": "2026-06-20T18:00:00+07:00",
    "created_at": "2026-05-30T16:56:08+07:00",
    "updated_at": "2026-05-30T16:56:08+07:00"
  }
}
```

POSSIBLE ERRORS

Error code	When it happens
bad_request	Missing tag or couple_name, or invalid JSON
create_failed	Duplicate tag (already exists)

2. List events

GET /api/events

Return all events, newest first.

RESPONSE EXAMPLE

```
{
  "status": "success",
  "data": [ { "id": 1, "tag": "stanley-arum", ... } ]
}
```

3. Get event detail

GET /api/events/:id

Return a single event by numeric id.

RESPONSE EXAMPLE

```
{ "status": "success", "data": { "id": 1, "tag": "stanley-arum", ... } }
```

POSSIBLE ERRORS

Error code	When it happens
event_not_found	No event with that id

Invitations

4. Import guests (CSV)

POST /api/invitations/import-csv

Upload a CSV of guests. Sent as **multipart/form-data** with a single file field named file. The event referenced by each row's **tag** must already exist. Each row is processed independently — bad rows are reported, they do not abort the import. An invitation_code and qr_code_token are generated automatically.

REQUEST BODY

Field	Type	Required	Description
file	file	yes	CSV with header: tag,guest_name,whatsapp_number

REQUEST EXAMPLE

```
# multipart/form-data, field "file"
tag,guest_name,whatsapp_number
stanley-arum,Budi Santoso,6281234567890
stanley-arum,Sinta Wijaya,6289876543210
```

RESPONSE EXAMPLE

```
{
  "status": "success",
  "message": "Import completed",
  "data": {
    "total_rows": 3,
    "success_count": 2,
    "failed_count": 1,
    "failed_rows": [
      { "row": 4, "reason": "event not found for tag",
        "tag": "kevin-michelle", "guest_name": "Budi Santoso" }
    ]
  }
}
```

POSSIBLE ERRORS

Error code	When it happens
bad_request	Missing file field, or empty / unreadable CSV

Note: HTTP 200 is returned even when some rows fail; inspect failed_count / failed_rows to surface per-row problems to the user. A row fails on: missing column, missing field, unknown tag, or a duplicate guest (same whatsapp_number already in that event).

5. List invitations

GET /api/invitations

Return invitations, newest first. All query params are optional and combine with AND. Each item embeds its parent event object.

QUERY PARAMETERS

Field	Type	Required	Description
tag	string	no	Filter by the event's tag
event_id	number	no	Filter by numeric event id
rsvp_status	string	no	not_answered attending not_attending
invitation_statuses	string	no	imported invitation_sent invitation_failed
qr_status	string	no	not_generated generated sent send_failed

REQUEST EXAMPLE

```
GET /api/invitations?tag=stanley-arum&rsvp_status=attending
```

RESPONSE EXAMPLE

```
{
  "status": "success",
  "data": [
    {
      "id": 1,
      "event_id": 1,
      "guest_name": "Budi Santoso",
      "whatsapp_number": "6281234567890",
      "invitation_code": "inv_7447ed7ae15f97d4",
      "invitation_status": "imported",
      "rsvp_status": "not_answered",
      "pax_count": null,
      "event_choice": null,
      "gift_interest": "not_asked",
      "qr_code_token": "0cc04bab-a900-4e9f-9644-69ce1da41841",
      "qr_status": "not_generated",
      "qr_sent_at": null,
      "created_at": "2026-05-30T16:56:08+07:00",
      "event": { "id": 1, "tag": "stanley-arum", ... }
    }
  ]
}
```

6. Get invitation detail

GET `/api/invitations/:id`

Return a single invitation by numeric id, including its event and any check-in logs.

POSSIBLE ERRORS

Error code	When it happens
invitation_not_found	No invitation with that id

WhatsApp send operations

These endpoints trigger outbound WhatsApp messages and update status fields. They return a **batch summary** rather than throwing on individual failures, so a partial failure still returns HTTP 200 — inspect the per-id results array.

Batch summary shape

```
{
  "status": "success",
  "message": "Batch processed",
  "data": {
    "total": 2,
    "succeeded": 2,
    "failed": 0,
    "results": [
      { "id": 1, "success": true },
      { "id": 2, "success": false, "reason": "..." }
    ]
  }
}
```

7. Send invitations (by id)

POST /api/invitations/send

Send the initial invitation to a specific set of invitations. Updates `invitation_status` to `invitation_sent` / `invitation_failed`.

REQUEST BODY

Field	Type	Required	Description
ids	number[]	yes	Invitation ids (at least one)

REQUEST EXAMPLE

```
{ "ids": [1, 2, 3] }
```

RESPONSE EXAMPLE

```
{ "status": "success", "message": "Invitations processed", "data": { ... } }
```

POSSIBLE ERRORS

Error code	When it happens
bad_request	Empty or missing ids array

8. Send pending invitations

POST /api/invitations/send-pending

Send the initial invitation to **all** guests of an event whose `invitation_status` = `imported`.

REQUEST BODY

Field	Type	Required	Description
tag	string	yes	Event tag

REQUEST EXAMPLE

```
{ "tag": "stanley-arum" }
```

9. Resend to unanswered

POST /api/invitations/resend-unanswered

Resend the invitation to all guests of an event whose `rsvp_status = not_answered`.

REQUEST BODY

Field	Type	Required	Description
tag	string	yes	Event tag

REQUEST EXAMPLE

```
{ "tag": "stanley-arum" }
```

10. Send reminder

POST /api/invitations/send-reminder

Send a reminder to all attending guests of an event.

REQUEST BODY

Field	Type	Required	Description
tag	string	yes	Event tag

REQUEST EXAMPLE

```
{ "tag": "stanley-arum" }
```

11. Generate & send QR

POST /api/invitations/generate-send-qr

Generate, upload and send the QR code to all eligible guests of an event. A guest is eligible when: `rsvp_status = attending`, `pax_count` is set, `event_choice` is set, and the QR has not been sent yet. Sets `qr_status = sent` and `qr_sent_at`.

REQUEST BODY

Field	Type	Required	Description
tag	string	yes	Event tag

REQUEST EXAMPLE

```
{ "tag": "stanley-arum" }
```

If no guests are eligible, total is 0 and results is empty — this is still a success response, not an error.

12. Resend QR (by id)

POST `/api/invitations/resend-qr`

Manually resend the QR to selected invitations within an event. If the QR media is missing it is regenerated and re-uploaded first. Ids that do not belong to the given tag are reported as failed in the results array.

REQUEST BODY

Field	Type	Required	Description
tag	string	yes	Event tag (guard)
ids	number[]	yes	Invitation ids to resend to

REQUEST EXAMPLE

```
{ "tag": "stanley-arum", "ids": [1, 2, 3] }
```

WhatsApp webhook

These endpoints are called by Meta, not by the front end. They are documented here for completeness.

13. Verify webhook

GET `/api/webhook/whatsapp`

Meta verification handshake. Echoes `hub.challenge` as plain text when `hub.verify_token` matches the server's configured token.

QUERY PARAMETERS

Field	Type	Required	Description
<code>hub.mode</code>	string	yes	Always "subscribe"
<code>hub.verify_token</code>	string	yes	Must match <code>META_VERIFY_TOKEN</code>
<code>hub.challenge</code>	string	yes	Echoed back on success

REQUEST EXAMPLE

```
GET /api/webhook/whatsapp?hub.mode=subscribe&hub.verify_token=...&hub.challenge=12345
```

RESPONSE EXAMPLE

```
200 OK
12345      # plain text, not JSON
```

POSSIBLE ERRORS

Error code	When it happens
<code>verification_failed</code>	Token did not match (403)

14. Receive webhook

POST `/api/webhook/whatsapp`

Receives inbound WhatsApp events from Meta and acknowledges with 200 OK. Payload parsing of interactive replies is handled server-side.

RESPONSE EXAMPLE

```
{ "status": "received" }
```

Check-in

These are the endpoints the scanner/check-in UI calls. After scanning a QR, the encoded URL contains the `qr_code_token`; use it to look up the guest, then POST the check-in.

15. Look up guest by QR

GET `/api/check-in/:qr_code_token`

Resolve a QR token to guest details and per-event check-in status. The `checked_in_events` array only includes events the guest's `event_choice` covers; `checked_in_at` is null until they check in for that event.

RESPONSE EXAMPLE

```
{
  "status": "success",
  "data": {
    "guest_name": "Budi Santoso",
    "pax_count": 2,
    "event_choice": "both",
    "checked_in_events": [
      { "event_type": "holy_matrimony", "checked_in_at": null },
      { "event_type": "reception",
        "checked_in_at": "2026-05-30T16:56:44+07:00" }
    ]
  }
}
```

POSSIBLE ERRORS

Error code	When it happens
invitation_not_found	No invitation for that QR token (404)

16. Check in

POST `/api/check-in`

Record a check-in for one event. Validated against RSVP status, event choice, pax count, and duplicate check-in. See the validation rules below.

REQUEST BODY

Field	Type	Required	Description
qr_code_token	string	yes	From the scanned QR
event_type	string	yes	holy_matrimony reception
actual_pax	number	yes	People arriving now; 1 .. pax_count
scanner_name	string	no	Operator name, for the audit log

REQUEST EXAMPLE

```
{
  "qr_code_token": "0cc04bab-a900-4e9f-9644-69ce1da41841",
  "event_type": "reception",
  "actual_pax": 2,
  "scanner_name": "Admin 1"
}
```

RESPONSE EXAMPLE

```
// 200 OK
{
  "status": "success",
  "message": "Check-in berhasil",
  "data": {
    "name": "Budi Santoso",
    "event_type": "reception",
    "registered_pax": 2,
    "checked_in_pax": 2
  }
}
```

POSSIBLE ERRORS

Error code	When it happens
invitation_not_found	Unknown QR token (404)
not_attending	rsvp_status is not "attending" (422)
invalid_event	event_choice does not allow this event_type (422)
invalid_pax	actual_pax is 0 or greater than pax_count (422)
already_checked_in	Guest already checked in for this event_type (409)

Check-in validation rules

Check	Rule
1. Token	Invitation must exist for the qr_code_token, else invalid_not_found
2. RSVP	rsvp_status must be "attending"
3. Event match	holy_matrimony → only holy_matrimony; reception → only reception; both → either
4. Pax	$1 \leq \text{actual_pax} \leq \text{pax_count}$
5. Uniqueness	Only one check-in per (invitation, event_type)

Recommended UI flow: scan QR → GET /api/check-in/{token} to show the guest and which events are still open → let the operator pick the event and enter actual_pax → POST /api/check-in. Map the returned error code to a clear on-screen message (e.g. already_checked_in → "Already checked in for this event").

Error code index

Complete list of machine-readable error codes the front end may receive.

Error code	HTTP	Meaning
bad_request	400	Validation / malformed body / bad multipart
event_not_found	404	GET /api/events/:id with unknown id
create_failed	409	Duplicate event tag on create
invitation_not_found	404	Unknown invitation id or QR token
verification_failed	403	Webhook verify token mismatch
not_attending	422	Check-in: guest not attending
invalid_event	422	Check-in: event_choice mismatch
invalid_pax	422	Check-in: actual_pax out of range
already_checked_in	409	Check-in: duplicate for event_type
internal_error	500	Unexpected server error

Project Vows API · v1.0 · generated for the front-end team. Datetimes are RFC3339 with timezone offset. All request/response bodies are application/json except CSV import (multipart) and the webhook verify response (plain text).